

The university database

Exercise of Entity-relationship modelling university database

The university database stores details about university students, courses, the semester a student took a particular course (and his mark and grade if he completed it), and what degree program each student is enrolled in.

As part of the conceptual design, create the entity relationship diagram from the following requirements list.

Use Peter Chen notation (there are no foreign keys on ERD)

- • A program is made up of one or more courses.
- • A student must enroll in a program.
- • A student takes the courses that are part of his program.
- • A program has a name, a program identifier, the total credit points required to graduate, and the year it commenced.
- A course has a name, a course identifier, a credit point value, and the year it commenced.
- A student has a given names, a surname, a student identifier, a date of birth, and the year they first enrolled. We can treat all given names as a single object—for example, “John Paul.”
- When a student takes a course, the year and semester he attempted it are recorded. When he finishes the course, a grade (such as A or B) and a mark (such as 60 percent) are recorded.
- Each course in a program is sequenced into a year (for example, year 1) and a semester (for example, semester 1).

The flight database (SU22 – 1605)

Exercise of Entity-relationship modelling flight database

The flight database stores details about an airline's fleet, flights, and seat bookings.

As part of the conceptual design, create the entity relationship diagram from the following requirements list.

Use Peter Chen notation (there are no foreign keys on ERD)

- An airplane has a model number, a unique registration number, and the capacity (to take passengers).
- An airplane flight has a unique flight number, a departure airport, a destination airport, a departure date and time, and an arrival date and time.
 - Each flight is carried out by a single airplane.
 - A passenger has given names, a surname, and a unique email address.
 - A passenger can book a reservation (with a booknumber, class, seatnumber, price, booktime, bookdate) on a flight.

The music database

The music database stores details of a personal music library, and could be used to manage your MP3, CD, or vinyl collection. Because this database is for a personal collection, it's relatively simple and stores only the relationships between artists, albums, and tracks. It ignores the requirements of many music genres, making it most useful for storing popular music and less useful for storing jazz or classical music

We first draw up a clear list of requirements for our database:

- o The collection consists of albums

- o An album is made by exactly one artist.

- o An artist makes one or more albums.

- o An album contains one or more tracks

- o Artists, albums, and tracks each have a name.

- o Each track is on exactly one album.

- o Each track has a time length, measured in seconds.

- o When a track is played, the date and time the playback began (to the nearest second) should be recorded; this is used for reporting when a track was last played, as well as the number of times music by an artist, from an album, or a track has been played There's no requirement to capture composers, group members or sidemen, recording date or location, the source media, or any other details of artists, albums, or tracks

